

Show Your Work

Quiz 7 Dec 4th 2025 Chem 1127

Name (First, Last):

Key

Note the quiz is front & back + Equation Sheet/Periodic Table

Points: ____/1620

1. The pressure of a sample of gas is measured at sea level ($P_{\text{atm}} = 760 \text{ mm Hg}$) with an open-end Hg (mercury) manometer (shown here). If $h = 30 \text{ cm}$, determine the pressure of the gas in mm Hg. (1 pt)

gas + Hg counters P_{atm}

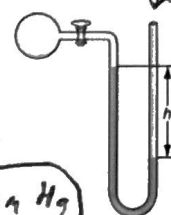
$$P_{\text{gas}} + P_{\text{Hg}} = P_{\text{atm}} \rightarrow P_{\text{gas}} = P_{\text{atm}} - P_{\text{Hg}}$$

$$(760 - 300) \text{ mm Hg}$$

- a. What has greater pressure, the atmosphere or the gas? Circle it (0.5 pt)

460 mm Hg

- b. Identify where in the manometer the atmosphere is. (0.5 pt)



2. Write the ideal gas law. (1.5 pt)

$$PV = nRT$$

- a. This semester only a single experiment was not done in lecture. It involved gas, explain how the gas's properties in the bottle changed, using the variables of the ideal gas law, and explain how it gave rise to the phenomenon that was observed between the egg & bottle. (1.5 pt)

to create less pressure in bottle, n must have decreased as combustion continued, sucking egg into bottle

3. H_2 gas is being considered for use as an alternative automotive fuel to replace gasoline. One gallon of gasoline could be replaced by 1 kg H_2 . What is the volume (in L) of this much H_2 at 25°C and 0.980 atm ? (2 pt)

$$\frac{1000 \text{ g H}_2}{1} \cdot \frac{1 \text{ mol H}_2}{2 \text{ g H}_2} = 500 \text{ mol H}_2$$

$$V = \frac{nRT}{P} = \frac{500 \text{ mol} \cdot 0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \cdot 298 \text{ K}}{0.980 \text{ atm}}$$

$$V = 1248 \text{ L}$$

4. What is the molar mass of a gas if 0.0494 g of the gas occupies a volume of 0.100 L at a temperature 26°C and a pressure of 0.404 atm ? (2 pt)

$$n = \frac{PV}{RT} = \frac{0.404 \text{ atm} \cdot 0.1 \text{ L}}{0.0821 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \cdot 299 \text{ K}} = 0.00165 \text{ mol}$$

$$MM = \frac{0.0494}{0.00165} = \frac{30 \text{ g}}{\text{mol}}$$

5. Briefly define the term diffusion. (1 pt)

Moving from High to Low concentration

6. Breaking chemical bonds is an endothermic/exothermic process, releasing/absorbing/requiring energy. Circle the correct bolded term in each pair. (1 pt)

or

Show Your Work

7. Identify the curve that corresponds to the gas at the highest temperature. briefly explain your answer (1 pt)

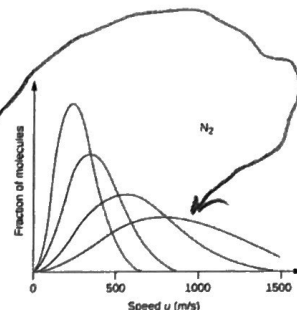
Farthest Right, gases travel fastest @

H.g. temp

- a. For two ideal gases with different molar mass at the same temperature, what is the same between them? Circle it below. (0.5 pt)

Average Kinetic Energy

Average Velocity



8. We discussed a theoretical space-based weapon involving heavy tungsten cylinders, colloquially called 'Rods from god'. Discuss how this weapon would have kinetic and potential energy. (1.5 pt)

Kinetic Energy

Potential Energy

Moving up to

Orbiting earth.

Space or gaining speed/Energy on descent

9. Consider if your hand is a system, you touch a red hot burner on your stovetop. Describe what the heat flow (q) is for your hand. (0.5 pt)

Heat flows into hand, q is positive

- a. Is the heat/energy solely used to raise your hand's temperature? Briefly discuss. (0.5 pt)

No, chemical bonds in Biological cells break, Ouch!

10. ~A12.27. In a coffee cup calorimeter, 0.800 g of ammonium nitrate (NH_4NO_3) is dissolved in 100.0 g of water. The water decreases in temperature from 21.11 to 20.39 °C. given that water has a specific heat capacity of $4.18 \text{ J g}^{-1} \text{ °C}^{-1}$, what is the amount of heat, in joules, transferred in this reaction? (1.5 pt)

$$q = \underset{\text{mass}}{100 \text{ g H}_2\text{O}} \cdot \underset{\text{specific Heat capacity}}{4.18 \frac{\text{J}}{\text{g} \cdot \text{°C}}} \cdot \underset{\Delta T}{(20.39 - 21.11) \text{ °C}} = -301 \text{ J}$$

11. On the given image of the bomb calorimeter, identify

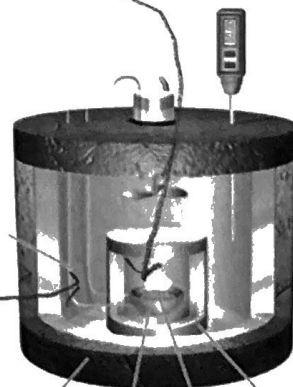
- a. where the sample is placed. (0.5 pt)

in center

- b. Whether the sample chamber has constant volume or pressure. (0.5 pt)

- c. Where a temperature change for the surroundings, is relevant to your experiment. You may briefly discuss this (0.5 pt)

in H_2O as this is measured



12. What is the density of fluorine (F_2) gas at STP? (1.5 pt)

1 mol occupies 22.4 L

MM of F_2 is

$$\text{MM} = \frac{38 \text{ g F}_2}{1 \text{ mol F}_2} \rightarrow 38 \text{ g}$$

$$\text{Density} = \frac{g}{L} = \frac{38 \text{ g F}_2}{22.4 \text{ L}} = 1.70 \text{ g/L}$$

- a. Assuming you have 1 mol of F_2 is a great move towards solving this problem. Fill in the blanks with the correct number and unit. Getting the main part right earns you this pt. Hint the value is the density of water. (0.5 pt)

Shortcut! could use $PV = nRT$ to ID this V too!